



KG COLLEGE OF ARTS AND SCIENCE

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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

STUDENTS ATTENDANCE MONITORING AND ELIGIBILITY VERIFICATION SYSTEM

Submitted by

BHARGAVI T R

25BIT103

Under the Guidance of

R MUGILAN

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled “**Students attendance monitoring and eligibility verification system**” submitted in partial fulfillment of the requirements for the course **B.Sc Information Technology** is a **record of my original work** carried out under the guidance of **R Mugilan**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Bhargavi T R

CERTIFICATE

This is to certify that the project entitled “**Students attendance monitoring and eligibility verification system**” is a **bona fide work** carried out by **Bhargavi T R , B.Sc Information Technology, KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Students Attendance Monitoring and Eligibility Verification System is a Java-based application designed to automate the process of tracking student attendance and determining their eligibility for examinations. In many educational institutions, attendance is recorded manually, which can lead to errors, time consumption, and difficulty in maintaining accurate records. This project aims to simplify and digitize the attendance management system.

The system allows users to record daily attendance of students, store the data efficiently, and calculate the overall attendance percentage automatically. Based on predefined criteria (such as minimum attendance requirement), the system verifies whether a student is eligible to appear for examinations. It reduces manual work and ensures accuracy in eligibility calculations.

The application is developed using Core Java concepts such as classes, objects, arrays, methods, loops, and conditional statements. It can be implemented as a console-based application or extended into a graphical user interface using Java Swing.

The main objective of this system is to improve efficiency in attendance management, minimize human errors, and provide quick and accurate eligibility reports. The system generates clear records showing student details, attendance percentage, and eligibility status.

This project demonstrates the practical application of object-oriented programming concepts in Java and provides a foundation for developing more advanced academic management systems in the future.

1. INTRODUCTION

In educational institutions, maintaining accurate attendance records is essential for monitoring student performance and ensuring academic discipline. Traditionally, attendance is recorded manually using registers, which is time-consuming and prone to errors. Managing large volumes of student data, calculating attendance percentages, and verifying eligibility for examinations manually can be difficult and inefficient.

With the advancement of computer technology, automated attendance systems have become important in improving accuracy and saving time. A computerized attendance monitoring system helps teachers record attendance quickly, maintain organized records, and automatically calculate attendance percentages.

The Students Attendance Monitoring and Eligibility Verification System is a Java-based application designed to automate the process of recording attendance and verifying student eligibility for examinations. The system allows the user to enter student details, mark daily attendance, and automatically calculate the attendance percentage. Based on predefined criteria (such as minimum attendance requirement), the system determines whether a student is eligible for exams.

This project is developed using Core Java programming concepts such as classes, objects, methods, arrays, loops, and conditional statements. The application can be implemented either as a console-based program or extended using Java Swing for a graphical user interface (GUI).

The primary goal of this system is to reduce manual effort, improve accuracy in attendance tracking, and provide quick eligibility results. This project also demonstrates how Java programming can be applied to develop real-world academic management systems.

1.1 Objective

The main objective of the Students Attendance Monitoring and Eligibility Verification System is to develop a simple and efficient application using Core Java that automates attendance tracking and eligibility checking in educational institutions.

The specific objectives of this project are:

- To design and develop a Java-based attendance monitoring system.
- To automate the process of recording and storing student attendance.
- To calculate attendance percentage automatically.
- To verify student eligibility based on minimum attendance criteria.
- To reduce manual errors in attendance and eligibility calculation.
- To provide a user-friendly system for teachers or administrators.

- To demonstrate the implementation of Core Java programming concepts such as classes, objects, methods, loops, and conditional statements.
- To generate clear reports showing attendance details and eligibility status.

1.2 Overview

The Students Attendance Monitoring and Eligibility Verification System is a menu-driven application developed using the Java programming language. The system provides a digital platform for recording and managing student attendance efficiently.

The system allows the user to enter student details and mark attendance on a daily basis. It stores attendance records and calculates the overall attendance percentage for each student automatically. Based on predefined rules (for example, 75% minimum attendance), the system determines whether the student is eligible to appear for examinations.

This system simplifies the attendance management process by eliminating manual calculations and reducing errors. The application can be developed as a console-based program or enhanced with a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in developing real-world applications for educational institutions.

1.3 Features

The Students Attendance Monitoring and Eligibility Verification System includes the following main features:

Student Record Management – Stores student details such as name, ID, and class.

Attendance Recording – Allows the user to mark daily attendance for students.

Automatic Attendance Calculation – Calculates attendance percentage automatically.

Eligibility Verification – Checks whether students meet the minimum attendance requirement.

Report Generation – Displays student attendance details and eligibility status.

User-Friendly Interface – Provides a simple and easy-to-use menu-driven system.

Data Accuracy – Reduces errors compared to manual attendance systems.

Time Saving – Speeds up attendance tracking and eligibility checking process.

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the system requirements are clearly identified. This phase helps in understanding how the system should function and what features are necessary to solve the problem effectively.

The analysis phase mainly focuses on identifying the drawbacks of the existing manual system and proposing a computerized solution that improves efficiency, accuracy, and reliability.

2.1 Problem Definition

In many educational institutions, student attendance is still recorded manually using registers. This traditional method creates several challenges.

Some common problems include:

- Errors in attendance recording
- Time-consuming manual calculations of attendance percentage
- Difficulty in maintaining and managing large student records
- Lack of accurate and organized data
- Difficulty in verifying student eligibility for examinations
- Increased workload for teachers and administrative staff

Manual attendance systems can also lead to incorrect eligibility decisions, which may affect students' academic progress.

Therefore, there is a need for an automated system that can efficiently record attendance, calculate attendance percentages, and verify student eligibility accurately.

The Students Attendance Monitoring and Eligibility Verification System is designed to solve these problems by providing a computerized solution that simplifies attendance management and eligibility checking.

2.2 Project Overview

The Students Attendance Monitoring and Eligibility Verification System is a Java-based application developed to simplify attendance tracking and eligibility verification in educational institutions.

The system provides a menu-driven interface where users can enter student details and record daily attendance. It automatically calculates the attendance percentage for each student and determines their eligibility based on predefined criteria (such as minimum attendance requirement).

The system includes the following functionalities:

- Recording student details
- Marking daily attendance
- Calculating attendance percentage automatically
- Verifying eligibility based on attendance criteria
- Generating reports showing attendance and eligibility status

This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application.

2.3 Scopes and Limitations

Scope

The scope of the Students Attendance Monitoring and Eligibility Verification System includes the following:

- Managing student records
- Recording and storing attendance data
- Calculating attendance percentage automatically
- Verifying student eligibility for examinations
- Generating attendance and eligibility reports
- Demonstrating the practical use of Java programming concepts

The system can be used in:

- Schools
- Colleges
- Universities
- Training institutes
- Coaching center

Limitations

The system may support only a limited number of students in basic versions

- It may not include advanced features like biometric or RFID attendance
- Data storage may be temporary in console-based applications
- Requires basic computer knowledge to operate
- Does not include online/cloud-based access in the basic implementation

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Students Attendance Monitoring and Eligibility Verification System is designed as a menu-driven application where users can enter student details, mark attendance, and verify eligibility automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the system are explained below:

1. Student Details module :

This module is responsible for storing student information such as:

- Student Name
- Student ID
- Class/Department

This data is used for maintaining proper attendance records.

Example Menu:

Student ID : 101

Name : Arun

Class:BCA

2. Attendance Entry Module:

This module allows the user to record daily attendance of students.

Functions of this module include:

- Marking attendance (Present/Absent)
- Accepting input for multiple student
- Storing attendance data

The system allows attendance to be recorded for each working day.

3. Attendance Calculation Module

This module performs the calculation of attendance percentage.

The system calculates:

- Total number of working days
- Total number of days present
- Attendance percentage

Example calculation :

$$\text{Attendance \%} = (\text{Days Present} / \text{Total Days}) \times 100$$

4. Eligibility Verification Module

This module checks whether a student is eligible for examination based on attendance.

- Minimum attendance criteria (e.g., 75%)
- If attendance $\geq 75\%$ $\rightarrow$ Eligible
- If attendance $< 75\%$ $\rightarrow$ Not Eligible

Example Output :

Attendance : 80%

Status : Eligible

5. Report Generation Module:

This module displays the final report for each student.

The report includes:

- Student Name & ID

- Total Days
- Days Present
- Attendance Percentage
- Eligibility Status

Example Output:

----- STUDENT REPORT -----

Name: Arun

ID: 101

Total Days: 100

Present: 80

Attendance: 80%

Status: Eligible

6. Exit Module

This module allows the user to exit the application safely after completing the process.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For this system, a structured development approach was followed to ensure efficiency and accuracy.

The development process includes the following stages:

- Problem Analysis

- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied in detail. The requirements of the system are identified based on educational needs.

The main problems identified in manual attendance systems are:

- Manual recording may lead to errors
- Time-consuming calculations
- Difficulty in maintaining records
- Hard to verify eligibility manually

Based on these issues, the system is designed to automate attendance and eligibility checking.

4.2. System Design

In this stage, the overall structure of the system is designed before coding.

The system is divided into modules such as:

- Student Details Module
- Attendance Entry Module
- Attendance Calculation Module
- Eligibility Verification Module
- Report Generation Module

- Exit Module

The program flow is designed to ensure easy user interaction.

4.3. Program Development

After designing the system, the program is implemented using Java.

The following steps were followed:

- Creating classes and methods
- Implementing menu-driven logic
- Writing code for attendance calculation
- Handling user input
- Displaying reports

Core Java concepts such as classes, objects, loops, arrays, and conditional statements are used.

4.4. Testing and Validation

Testing ensures that the program works correctly.

Example Test Cases:

Test Case	Expected Result
Enter student details	Data stored correctly
Mark attendance	Attendance recorded

Test Case	Expected Result
Calculate percentage	Correct result displayed
Check eligibility	Proper status shown

This phase ensures accuracy and error-free execution.

4.5. Result and Documentation

After testing, the results are documented.

The system generates:

- Attendance reports
- Eligibility status

Screenshots and outputs are used to demonstrate system working.

Proper documentation explains the design and implementation.

4.6. Algorithm

Algorithm for the system:

- 1.Start the program
- 2.Enter student details
- 3.Input total working days
- 4.Enter days present

5. Calculate attendance percentage

6. Check eligibility ($\geq 75\%$)

7. Display student report

8. End the program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start



Enter Student Details



Enter Attendance



Calculate Percentage



Check Eligibility



Display Report



End

5.SOURCE CODE

```
import java.util.Scanner;

class Student {
    int id;
    String name;
    int totalDays;
    int presentDays;
    double percentage;
    String status;

    // Method to input student details
    void inputDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student ID: ");
        id = sc.nextInt();
        sc.nextLine(); // clear buffer

        System.out.print("Enter Student Name: ");
        name = sc.nextLine();

        System.out.print("Enter Total Working Days: ");
        totalDays = sc.nextInt();

        System.out.print("Enter Days Present: ");
        presentDays = sc.nextInt();
    }
}
```

```
}
```

```
// Method to calculate attendance percentage
```

```
void calculateAttendance() {
```

```
    percentage = ((double) presentDays / totalDays) * 100;
```

```
}
```

```
// Method to check eligibility
```

```
void checkEligibility() {
```

```
    if (percentage >= 75) {
```

```
        status = "Eligible";
```

```
    } else {
```

```
        status = "Not Eligible";
```

```
    }
```

```
}
```

```
// Method to display report
```

```
void displayReport() {
```

```
    System.out.println("\n----- STUDENT REPORT -----");
```

```
    System.out.println("Student ID   : " + id);
```

```
    System.out.println("Name       : " + name);
```

```
    System.out.println("Total Days  : " + totalDays);
```

```
    System.out.println("Present Days : " + presentDays);
```

```
    System.out.printf("Attendance   : %.2f%%\n", percentage);
```

```
    System.out.println("Status      : " + status);
```

```
}
```

```
}
```

```
public class AttendanceSystem {
```

```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of students: ");
    int n = sc.nextInt();

    Student[] s = new Student[n];

    // Input details for each student
    for (int i = 0; i < n; i++) {
        System.out.println("\nEnter details for Student " + (i + 1));
        s[i] = new Student();
        s[i].inputDetails();
        s[i].calculateAttendance();
        s[i].checkEligibility();
    }

    // Display reports
    System.out.println("\n===== STUDENT ATTENDANCE REPORT =====");
    for (int i = 0; i < n; i++) {
        s[i].displayReport();
    }

    System.out.println("\nProgram Ended Successfully!");
}
}
```

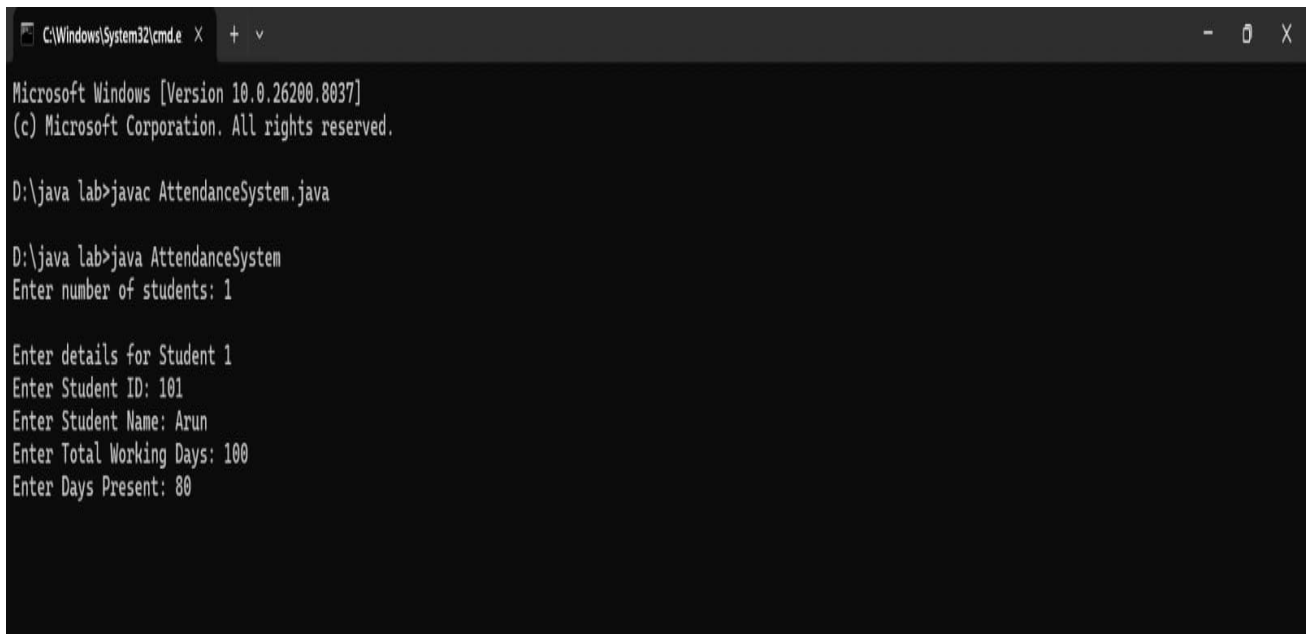
6.INPUT AND OUTPUT SCREENS



```
C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\java lab>javac AttendanceSystem.java

D:\java lab>java AttendanceSystem
```



```
C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\java lab>javac AttendanceSystem.java

D:\java lab>java AttendanceSystem
Enter number of students: 1

Enter details for Student 1
Enter Student ID: 101
Enter Student Name: Arun
Enter Total Working Days: 100
Enter Days Present: 80
```

```
C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\java lab>javac AttendanceSystem.java

D:\java lab>java AttendanceSystem
Enter number of students: 1

Enter details for Student 1
Enter Student ID: 101
Enter Student Name: Arun
Enter Total Working Days: 100
Enter Days Present: 80

===== STUDENT ATTENDANCE REPORT =====

----- STUDENT REPORT -----
Student ID   : 101
Name        : Arun
Total Days   : 100
Present Days : 80
Attendance   : 80.00%
Status       : Eligible

Program Ended Successfully!

D:\java lab>
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Students Attendance Monitoring and Eligibility Verification System was developed to automate the process of recording student attendance and verifying their eligibility for examinations. The system allows users to enter student details, record attendance, calculate attendance percentage, and determine eligibility automatically.

During the development process, Core Java programming concepts such as loops, conditional statements, arrays, methods, and object-oriented principles were applied. The system successfully calculates attendance percentages and accurately determines whether a student meets the minimum attendance requirement.

The system was tested with different inputs to ensure that attendance calculations and eligibility verification are performed correctly. The results show that the system reduces manual effort, minimizes errors, and improves efficiency in attendance management.

This project demonstrates how a simple Java-based application can effectively replace manual attendance systems and provide accurate and reliable results.

7.2 Conclusion

The Students Attendance Monitoring and Eligibility Verification System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world academic application.

The system automates attendance tracking, reduces errors in calculation, and provides quick eligibility results. It simplifies the process of managing student attendance and ensures that eligibility is determined accurately based on predefined criteria.

This project helps students understand how Java programming can be used to develop practical software systems for educational institutions. The system can be further enhanced by adding advanced features such as database integration, graphical user interface (GUI), biometric attendance, and online access.

8.REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*
- E. Balagurusamy – Programming with java

Websites

1. <https://www.oracle.com/java>
2. <https://www.geeksforgeeks.org>
3. <https://www.w3schools.com>
4. <https://www.javatpoint.com>
5. <https://www.tutorialspoint.com/java>